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August 6, 1979
SUPERSEDING
Int. Fed. Spec. P-S-00619B(GSA-FSS)
August 1, 1975 and
Fed. Spec. P-S-619A
June 17, 1974

FEDERAL SPECIFICATION

SOAP, ANTIMICROBIAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two grades of cake toilet soap containing antimicrobial agents (see 6.1).

1.2 Classification. Antimicrobial soap shall be in cake form and of the following grades, styles, and sizes, as specified (see 6.2).

Grade A - Broad spectrum.
Grade B - Regular.

Style 1 - Wrapped cake.
Style 2 - Unwrapped cake.

Size 1 - 1/2 to 1 ounce.
Size 2 - 1-1/2 to 2 ounces.
Size 3 - 2-1/2 to 3-1/2 ounces.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 1 - General Regulations for the Enforcement of the Federal Food, Drug, and Cosmetic Act and the Fair Packaging and Labeling Act.
43 FR 4, pp 1243-1244 - OTC Topical Antimicrobial Products, "Test for Antimicrobial Soap Effectiveness."

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Association, Inc., Agent:

National Motor Freight Classification

Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standard:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Finished product. The finished product shall be a thoroughly saponified cake of soap containing an antimicrobial agent(s). The soap cakes shall be mildly perfumed. The soap cakes shall be milled, uniform in size, firm and well compressed. The surfaces and edges of the cakes shall be smooth. The cakes shall be free from foreign matter, cracks, and chips.

3.1.1 The finished product shall not contain rosin, when tested as specified in 4.3.4.

TABLE I. Physical and chemical requirements

Property	Requirement	
	Minimum	Maximum
Moisture and matter volatile at 105°C, percent	---	15.0
Free alkali, calculated as sodium hydroxide (NaOH), percent	---	0.1
Matter insoluble in water, percent	---	1.0
Anhydrous soap, percent	79.0	---
Acid Number, mg KOH/g mixed fatty acids from soap	205	---

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3.2 Physical and chemical requirements. The finished product shall conform to the requirements of table I, when tested as specified in 4.3.3.

3.3 Cake weight. The cakes shall be uniform in weight. The manufacturer shall state in the contract what weight cakes are offered for each size. The average weight of the cakes supplied under the contract shall be not less than the weight specified in the contract (see 6.3).

3.4 Active ingredient. When tested as specified in 4.3.5, the soap shall contain one or more antimicrobial ingredients at a concentration which has been demonstrated to pass the requirements of 3.4.1 and 3.4.2.

3.4.1 In-vitro bacterial growth inhibition. When tested as specified in 4.3.1, grade A antimicrobial soap shall completely inhibit the growth of Staphlococcus aureus (ATCC No. 6538), Escherichia coli (ATCC No. 11229), Proteus mirabilis (ATCC No. 9921), and Salmonella typhimurinum (ATCC No. 13311). Grade B antimicrobial soap shall completely inhibit the growth of Staphlococcus aureus (ATCC No. 6538).

3.4.2 In-vivo bacterial growth inhibition. When tested as specified in 4.3.2, both grades of antimicrobial soap shall reduce the number of bacteria on the hands a minimum of 70 percent.

3.5 Federal Food, Drug and Cosmetic Act. The product shall comply with the Federal Food, Drug and Cosmetic Act, and regulations published under the Act.

3.6 Wrapping. Style 1, wrapped cakes, shall be completely wrapped in accordance with commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Quality conformance inspection.

4.2.1 Examination of the soap. The soap shall be examined for compliance with 3.1. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-4, acceptable quality level (AQL) 2.5 percent defective. The sample unit shall be one cake of soap.

4.2.2 Examination of the soap cakes for net weight. The soap cakes shall be examined for compliance with 3.3. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-4, AQL 4.0 percent defective. The sample unit shall be one box of soap cakes. Determine the average weight of the cakes in the box.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, AQL 4.0 expressed in terms of percent defective. The sample unit shall be one shipping container fully prepared for delivery.

4.2.4 Testing of the soap. The soap shall be tested as specified in 4.3.3 - 4.3.5 for compliance with the requirements of section 3. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-1, AQL of 4.0 percent defective. The sample unit shall be one cake of soap.

4.3 Test methods.

4.3.1 In vitro bacterial growth inhibition.

4.3.1.1 Bacterial cultures. Maintain the bacterial strains on trypticase soy agar slants (Baltimore Biological Laboratories) or equivalent. Store in a refrigerator at 6° - 10° C. Transfer the cultures at intervals not exceeding 14 days. For the bacterial growth inhibition test, make three serial 24 hour subcultures of the bacterial strains in tubes of trypticase soy broth maintained at 37°C. Use the final serial subculture for the inhibition test.

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4.3.1.2 Preparation of culture plates. Place a layer of tryptone glucose extract agar containing 0.001 gram of antimicrobial soap per milliliter of agar solution in test culture plates. Place a layer of tryptone glucose extract agar containing 0.001 gram of a nonantimicrobial soap such as "Ivory" per milliliter of agar solution in control culture plates. Do not heat solutions containing soap above 50°C. Incubate the plates at 37°C for 48 hours. Discard any culture plates that were contaminated and supported bacterial growth.

4.3.1.3 Inhibition test. Inoculate two sterile test culture plates and two sterile control culture plates with 0.1 ml of the bacterial strain against which the antimicrobial soap is being tested (see 3.4.1). Each strain will be tested separately. Incubate the plates for 48 hours at 37°C. No bacterial growth shall occur on the test culture plates. Viability of the culture must be demonstrated by bacterial growth on the control plates.

4.3.2 In-vivo bacterial growth inhibition. The antimicrobial soap shall be tested as specified in 43 CFR 4, pages 1243, 1244 - "Test for Antimicrobial Soap Effectiveness".

4.3.3 Analysis of the soap cakes. The soap cakes shall be analyzed by the referenced sections of ASTM D 460 for the requirements of table I.

4.3.3.1 Moisture - section 11.

4.3.3.2 Free alkali (as NaOH) - section 18.

4.3.3.3 Matter insoluble in water - section 19.

4.3.3.4 Anhydrous soap - sections 21 and 22.

4.3.3.5 Acid number of fatty acids - sections 45 and 46.

4.3.4 Qualitative test for rosin (Liebermann-Storch). Transfer 1 to 2 milliliters of the fatty acids prepared in section 42 of ASTM Standard D 460 to a test tube, treat with 5 to 10 milliliters of reagent grade acetic anhydride, and warm on a steam bath. After cooling, pour 1 to 2 milliliters into a white porcelain dish and allow a drop or two of sulfuric acid (sp. gr. 1.53) to run down the side of the vessel. (Sulfuric acid of sp. gr. 1.53 may be prepared by adding 34.7 ml of sulfuric acid (sp. gr. 1.84) to 37.5 ml of distilled water.) If rosin is present, a fugitive violet coloration changing to a brownish tinge is immediately produced at the margin of contact of the reagents. This test can be made on a smaller sample on a spot plate. The tests should be checked with a sample of pure stearic acid to which 3 percent of rosin has been added.

4.3.5 Active ingredient. Analyze the antimicrobial soap by the following methods for conformance to the manufacturer's formulation which has passed the requirements of 3.4.1 and 3.4.2. The manufacturer shall furnish the formulation and test results to the quality assurance specialist.

4.3.5.1 Triclosan. Weigh approximately 1 g of soap shavings to the nearest mg (W_o, in g), and stir with 10 ml of distilled water in a 50 ml beaker until a homogeneous slurry is obtained. Transfer quantitatively to a separatory funnel. Make the slurry acidic to litmus paper with sulfuric acid solution. Add 100 ml methylene chloride and 20 ml of a 20 percent aqueous sodium sulfate solution. Shake for 2 minutes and allow to stand until complete separation has occurred. Transfer the organic phase to a 250 ml beaker. Extract again as above with 100 ml methylene chloride and 20 ml 20 percent sodium sulfate solution. Transfer the organic phase to the 250 ml beaker. Place the beaker in a vacuum oven at 60°C until the solvent has evaporated off. Take up the residue in 10 ml of ethanol. Weigh approximately 0.02 g of triclosan to the nearest 0.1 mg (W_s, in g) and dissolve in 10 ml of ethanol to make a standard triclosan solution. Analyze the standard and unknown in a temperature-programmable gas chromatograph (eg: Hewlett-Packard Model 5750; carrier gas - helium at 35 ml/min; injection port temperature 270°C; column temperature 220° - 325° C at 10°C/sec, then isothermal; detector temperature 340°C; ECD and FID dual channel detectors; glass column, 6 foot length, 4 mm inside diameter, column packing SV2250 (Supelco, Inc.); 5 microliter samples injected). Based on the areas under the triclosan peaks in the standard (A_s) and unknown (A_u), calculate the percent triclosan in the soap.

$$\text{Percent triclosan} = \frac{A_u \times W_s \times 100}{A_s \times W_o}$$

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NOTE: Ethanol has a high coefficient of volume expansion with temperature. For accurate results, all ethanol, ethanol solutions and glassware, including pipets and cells should be at room temperature. Cooling due to ethanol evaporation, and warming due to handling should be kept to a minimum.

4.3.5.2 Triclocarban. Weigh approximately 1 g of soap shavings to the nearest mg (W_o, in g) and place in a 50 ml beaker. Add 10 ml of dimethylformamide and mix thoroughly. Filter under suction through a glass-frit filter. Evaporate 3.0 ml of the filtrate to dryness on a steam-bath or hotplate. Take up the residue in 10-20 ml ethanol and transfer quantitatively to a 100 ml volumetric flask. Fill to the mark with ethanol and mix well. Transfer 4.0 ml of this solution to another 100 ml volumetric flask. Add 4.0 ml of concentrated ammonium hydroxide, fill to the mark with ethanol and mix well. Weigh approximately 0.04 g triclocarban to the nearest 0.1 mg (W_s, in g), dissolve in ethanol and transfer quantitatively to a 1 liter volumetric flask. Fill to the mark with ethanol and mix well. Transfer 4.0 ml of this solution to a 100 ml volumetric flask, add 4.0 ml concentrated ammonium hydroxide, fill to the mark with ethanol and mix well. Make a blank solution by adding 4.0 ml concentrated ammonium hydroxide to a 100 ml volumetric flask, filling to the mark with ethanol and mixing well. Determine the absorbance at 264 nm of the unknown (A_u), the standard (A_s) and the blank (A_b) with a spectrophotometer (eg: Carey Model No. 14). Calculate the percent triclocarban in the soap.

$$\text{Percent triclocarban} = \frac{(A_u - A_b) \times W_s \times 100}{3 \times (A_s - A_b) \times W_o}$$

CAUTION: See note to 4.3.5.1. Prepare all solutions fresh each day.

When the soap contains both triclosan and triclocarban, prepare a standard solution of triclosan. Weigh approximately 0.04 g of triclosan to the nearest 0.1 mg (W_{s1}, in g), dissolve in ethanol, transfer quantitatively to a 1 liter volumetric flask, fill to the mark with ethanol and mix well. Add 4.0 ml of this solution to a 100 ml volumetric flask, add 4.0 ml of concentrated ammonium hydroxide, fill to the mark with ethanol and mix well. Determine the absorbance of this triclosan solution (A_{s1}) at 264 nm when the other determinations are made. The expected absorbance due to the triclosan in the soap is given by:

$$A_e = \frac{3 \times W_n \times (Y/100) \times (A_{s1} - A_b).}{W_{s1}}$$

Where Y is the percent triclosan in the soap as determined in 4.3.5.1. The percent triclocarban in a soap containing both triclosan and triclocarban is given by:

$$\text{Percent triclocarban} = \frac{(A_u - A_e - A_b) \times W_s \times 100}{3 \times (A_s - A_b) \times W_o}$$

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Cakes of soap of one grade, style and size shall be packed in quantities specified in table II in a box conforming to PPP-B-636, class weather resistant. The unwrapped soap cakes shall have protective pads of paper or paperboard covering the entire top and bottom area of the pack. The box shall be closed, waterproofed and banded in accordance with the box specification.

TABLE II. Number of cakes per shipping container

Size	Number of cakes
1	1000
2	500
3	200

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5.1.2 Commercial. Cakes of soap of one grade, style, and size and in quantity specified shall be packed in a fiberboard box to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.2 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The antimicrobial soap covered by this specification is intended for use with water to clean the skin where a temporary reduction of the total number of bacteria on the skin is desired. Both grades of antimicrobial soap provided a temporary deodorant effect by reducing the number of gram positive bacteria on the skin. Grade 1 antimicrobial soap also possesses bacteriostatic activity against gram negative bacteria in vitro.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grade, style and size required (see 1.2).
(c) Packing required (see 5.1).
(d) Marking required (see 5.2).

6.3 Bid evaluation. Bids should be evaluated on the basis of price per ounce of soap cake offered. Bidders are asked to state the weight (in ounces) of the soap cake that they will furnish. The price per ounce of soap is determined by dividing the bid price per cake by the weight of the soap cake offered.

MILITARY INTEREST:

User Activity:

DPSC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
HEW - FDA
VA - DMS

Preparing Activity:

GSA - FSS

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